databricks

Keep the customers who are already telling us they will leave

Streamline Telco Save Desk | Churn early warning on the Databricks Data Intelligence Platform

Demo built on synthetic data. Streamline Telco is a fictional company.

Databricks account team · proposal for the Chief Customer Officer,
the Head of Retention and the Head of Data Platform · October 2026

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The outcome: one point less churn, €5M a year kept

What we propose, what it is worth and what it costs

Annual churn target

16.8% → 15.8%

-1.0 point in year one

Subscribers kept per year

20,000

out of ~336,000 who leave today

Revenue protected per year

€6.5M

€5.0M net of offers and platform

Time to first value

6 weeks

POC ≈ $1.6k compute · run ≈ $3.5k / month

For the Chief Customer Officer

- Churn falls and revenue stays, with a number you can track every month
- Every save is logged and compared with a control group, so the result is measured

For the Head of Retention

- Agents start each day with a ranked list: who to call, why, and what to offer
- Calling the model's top 10% reaches 3.7× more real churners than random outreach

Business figures are planning assumptions for Streamline (slide 10). Model figures are measured on the synthetic-data build (evidence/model_metrics.json).

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Today churn becomes visible only after the customer has left

The signals already exist, but they sit in three systems that nobody joins up



CARE CALLS

Customers say it out loud

"Match it or I am porting my number out." Stored as recordings and transcripts, never read at scale.



NETWORK

Experience degrades first

Dropped calls and slow data appear weeks before the porting request, but stay with the network team.



BILLING AND CONTRACT

Bill shock, contract end

A bill increase or a contract ending within 60 days opens the door to a competitor offer.

Today

- Retention reacts to the porting request, when the decision is already made
- Offers are generic and go to whoever calls
- Analysts wait days for BI tickets on "where and why"

With the Save Desk

- Every subscriber scored nightly, the riskiest and most valuable first
- The offer matches the reason: network fix, price match or early renewal
- Managers ask Genie in plain English and get the answer in seconds

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What we learned from the build

Results on 20,000 synthetic subscribers and 13,749 care calls, committed as run output in the repository

Churn after a switching mention

45.2%

vs 10.4% for everyone else: 4.3× higher

Precision of the top-10% list

56%

3.7× the 15.1% base rate

Value of reading calls with Gen AI

+0.053

ROC-AUC 0.729 → 0.783 · top-10% precision 42% → 56%

Main reason for at-risk customers

39 / 37 / 24%

price / network / contract

How we measured it: a gradient-boosting model with 21 features, tracked in **MLflow**, tested on a 25% hold-out. To isolate the value of Gen AI we retrained the same model without the features that **AI Functions** extract from care calls (an ablation). Switching intent is the #1 feature (importance 0.31).

> **The strongest single predictor is what the customer says on the care call. That signal is free; we only need to read it.**

Source: evidence/model_metrics.json, evidence/model_run_output.txt, evidence/query_results.md (synthetic data, seed 42).

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A day with the Save Desk

What changes for the people who own the outcome, and what the demo shows

1

02:00 · The platform does the reading

New bills, network data and yesterday's calls are ingested; AI reads every call; every subscriber gets a fresh score.

2

08:30 · The agent opens the save list

Ranked by value at risk, with the top three reasons, the next best offer and a suggested opening line.

3

On the call · Log the outcome

Accepted, declined or call back: one click, saved in milliseconds, used in next week's retraining.

4

Anytime · The manager asks Genie

"Which region has the most at-risk subscribers because of network problems?" The answer comes with its SQL.

LIVE DEMO · ABOUT 6 MINUTES

Tell, show, tell

- **Tell:** the retention team's morning, in their words
- **Show:** the top subscriber: 89% risk, €2.4k at stake, bill up 50%, mentioned switching. Offer: loyalty price match. Log "accepted".
- **Show:** filter to network-driven risk in the South, the list for the network team
- **Show:** ask Genie the switching question: 45% vs 10%
- **Tell:** what it means for churn, revenue and cost per save

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One governed journey from raw data to the agent's screen

Every stage runs on the Databricks Data Intelligence Platform and deploys as one Databricks Asset Bundle



Raw data

Billing, network probes, contracts, care-call transcripts



Ingest

Lakeflow

Declarative pipeline, Auto Loader, data-quality expectations



Govern

Unity Catalog

Medallion schemas, tags, column mask, grants, lineage



Intelligence

AI Functions · MLflow

Sentiment, intent, switch intent; churn model in UC



Serve

Lakebase

Save list synced to Postgres; outcomes written back



Ask

Genie

Plain-English questions on the same gold tables



Act

Databricks Apps

Save Desk for agents and managers

Integrated, not stitched

The app reads **Lakebase**, synced from the gold table the model writes, built from the calls AI enriched. One lineage graph in **Unity Catalog**.

Closed loop

Outcomes logged in the app return to the lakehouse. The model learns which offers actually save which customers.

Production path

Synthetic files are replaced by **Lakeflow Connect** or Kafka feeds from billing, network and the contact-centre platform. Nothing downstream changes.

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Security and governance by design

Transcripts are the sensitive asset, so they get the strictest controls

SECURITY

Data never leaves the governed platform

- **AI Functions** call Databricks-hosted models; transcripts are not sent to a third-party API
- Raw transcripts are tagged sensitive and masked by a **Unity Catalog** column mask. Only the data team sees the text.
- The retention team and **Genie** read gold tables only
- **Lakebase** uses the same workspace identity; the app runs as its own service principal

DATA QUALITY

Quality and trust

- **Lakeflow** expectations drop or flag bad rows (e.g. an implausible bill) before they reach the model
- LLM intent agrees with the IVR queue on 95% of calls. The POC adds a check against 500 calls labelled by people.
- Every score carries its top three reasons, so agents can trust and explain it
- **Genie** shows the SQL behind every answer; trusted queries are curated

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Decisions and trade-offs

Why this design, and what we deliberately did not do

Decision	Why	Trade-off we accept
AI Functions in SQL, not a fine-tuned model	Short transcripts, generic task; ships in days with no model to host	Per-token cost at scale. We process only new calls and can move to one combined prompt (about -60% tokens).
Gradient boosting, not deep learning	Tabular data; agents need the "why"; retrains in minutes	A more complex model might gain some AUC but lose explainability
Lakebase for the app, not Delta reads	Point lookups and write-back are transactional work that needs millisecond reads	A second store, kept in sync declaratively by a synced table
Rule-based next best offer first	Retention owns the offer catalogue; rules are easy to audit	Not optimised yet. Logged outcomes later train an uplift model.
Nightly batch scoring, not real time	Save decisions are daily; batch cuts cost and complexity	Same-day signals wait until the next run. Real time is a phase-3 option.

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Value, cost and plan

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Business case: about €5.0M net per year

Conservative planning assumptions, replaced by Streamline actuals in week 1 of the POC

Assumption	Value
Postpaid subscribers	2,000,000
Monthly churn (annual)	1.4% (16.8%)
Subscribers lost per year	336,000
Average revenue per subscriber	€27 / month
Revenue lost to churn per year	€108.9M
Precision of top-10% list (build)	56%
Save rate on reached churners	25%
Average offer cost (offer mix)	€40

Year-one result	Value
Subscribers saved (-1.0 pt churn)	20,000
Churners to reach (20,000 ÷ 25%)	80,000
Calls to make (80,000 ÷ 56%)	≈143,000 (≈550 / day)
Revenue protected (20,000 × €324)	€6.48M
Offer cost (≈35,600 accepted × €40)	-€1.42M
Platform run cost (slide 11)	-€0.04M
Net benefit per year	≈ €5.0M
Cost per save	≈ €73

Accepted offers = 20,000 saves + 25% of ≈62,600 contacted non-churners. Revenue view; margin view built with Finance in the POC. Streamline Telco is fictional; all business figures are illustrative.

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Cost: about $1.6k for the POC, $3.5k a month in production

Databricks list-price estimates (AWS, pay-as-you-go), confirmed with a Lakemeter sizing before any quote

Component	POC · 6 weeks · 200k subscribers	POC	Production · 2M subscribers	Per month
AI Functions on care calls	200k-call backfill + re-runs	$400	~690k calls, 3 AI functions per call	$1,170
Lakeflow pipeline	~150 DBU serverless	$60	Nightly incremental, ~600 DBU	$240
MLflow training + scoring	~200 DBU serverless	$80	Nightly scoring + weekly retrain, ~300 DBU	$120
SQL warehouse (Genie, SQL)	Small, ~3 h per working day	$760	Small, ~4 h / working day + 1 h nightly	$990
Lakebase	1 CU, working hours	$150	2 CU, always on	$730
Databricks Apps	Working hours (~300 h)	$105	Always on	$255
Cloud storage	< 100 GB	$5	~1 TB	$25
Total		≈ $1,560	≈ $42k per year	≈ $3,530

Unit prices: Foundation Model API, Llama 3.3 70B, $0.50 per 1M input and $1.50 per 1M output tokens (read on databricks.com/product/pricing/foundation-model-serving). Estimates to validate: serverless jobs and pipelines $0.40/DBU · serverless SQL $0.70/DBU (Small = 12 DBU/h) · Lakebase $0.50/DBU with 1 CU ≈ 1 DBU/h · App 0.5 DBU/h at $0.70. One combined AI prompt instead of three cuts production to about $2.8k a month.

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Plan: prove it in six weeks, then scale with a control group

Each phase has a go/no-go measured in the retention team's own KPIs

WEEKS 1–6 · POC

Prove the signal on real history

- 200k subscribers and 3 months of calls in a Streamline workspace
- Back-test: top-10% precision ≥ 2.5× base rate
- AI switch intent ≥ 85% agreement with 500 calls labelled by people
- Save Desk live for 10 agents

MONTHS 2–3 · PILOT

Prove the saves

- One region, 20 agents, a randomised control group
- Go if the save rate beats control by ≥ 5 points at ≤ €80 per save
- Production feeds via Lakeflow Connect

MONTH 4+ · SCALE

All regions, closed loop

- Every agent, nightly scores, weekly retraining on outcomes
- The network team gets its own at-risk list
- Next: uplift model for offers, real-time scoring

What we need from Streamline

An executive sponsor in the Customer Office · a retention lead who owns the offer catalogue and the KPI · one data engineer for the source feeds · approval to use 3 months of anonymised care-call transcripts in the POC workspace.

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Questions we expect, and our answers

Business and technical concerns

Business

- **Discounts for people who would stay?** Some, and it is priced in (€1.42M). The control group measures the real gain; a network fix costs €15, not a discount.
- **How fast is the value?** A back-tested list in 6 weeks; first measured saves in month 2.
- **What if the numbers are lower?** At half the save rate, net benefit is still ≈ €2.2M a year against ≈ $42k platform cost.

Technical

- **Integration?** **Lakeflow Connect** or Kafka from billing, network and contact centre; outcomes in Postgres, readable by the CRM.
- **Data quality?** Pipeline expectations, LLM labels checked against people, drift tracked in **MLflow**.
- **Security?** **Unity Catalog** masks, grants and lineage; Databricks-hosted models; app service principal.
- **Lock-in?** Delta and Postgres are open formats; the model is standard scikit-learn in MLflow.

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The decision we are asking for

Approve a six-week POC and name its owners

Time to a back-tested save list

6 weeks

on Streamline's own data

Platform compute for the POC

≈ $1.6k

list-price estimate

Net annual benefit at -1 pt churn

€5.0M

measured against a control group

Next steps

- Confirm the executive sponsor and the retention lead
- Share 3 months of anonymised data in a Streamline Databricks workspace
- Kick-off workshop: agree the KPI baseline, the control group and the offer catalogue

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Appendix: build evidence and references

Everything below is committed in the repository as text

Run evidence

- evidence/model_run_output.txt: full pipeline log (silver → Gen AI → features → model → save list → serving → Genie)
- evidence/model_metrics.json: ROC-AUC 0.783, PR-AUC 0.486, top-10% precision 0.562, ablation 0.729
- evidence/query_results.md: five Genie questions with SQL and answers
- data/output/*_sample.csv: raw and gold samples; app/demo_data.js: the app data

References

- Foundation Model API pricing: databricks.com/product/pricing/foundation-model-serving
- Databricks pricing: databricks.com/product/pricing
- AI Functions: docs.databricks.com/aws/en/large-language-models/ai-functions
- Lakebase synced tables: docs.databricks.com/aws/en/oltp/projects/sync-tables
- Genie: docs.databricks.com/aws/en/genie

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